Likely questions

Short, honest answers for the Q&A

Short, honest answers. If you don't know, say "we haven't measured that yet; here is what we did measure."

Is the data real?

News, IMD weather and warnings, CPCB air quality, the Flood Monitor, lake levels and market prices are real and collected every hour. Grievances, police, PWD and hospital records are simulated, because those departments have no open API. They share the real rain calendar and real wards, and every simulated row is flagged. Connecting a real department means writing one loader.

Why did you not use the two FarmwiseAI systems?

We explored both. The hostels system is for GHMC, which is Hyderabad, and the leave portal is for Yadadri Bhuvanagiri in Telangana. Neither has Chennai data. The same kind of source, a website with a login, plugs in through Data Studio.

How do you know two reports are the same incident?

We compare only plausible pairs: same category group, nearby, and close in time. A logistic regression scores each pair on distance, time gap, word and meaning similarity, shared names, category and ward. Pairs above a threshold learned from the data are merged; the grey zone goes to a review list. News must also share the category. F1 is about 0.83 on held-out simulated events.

How do you match Tamil and English news?

They rarely share words, so we match by meaning. bge-m3 puts both languages in one vector space; a cross-encoder reranks the candidates; a model trained on our labelled pairs decides; an LLM checks only the unsure pairs. We also prevent chaining.

How is priority decided? Is it AI?

No. Severity is a rule score from facts. Priority adds weighted, named signals: past deadline, more than one source, vulnerable people, rain, growing, recurring. Each reason is shown. The weights can learn from the Collector's decisions after 30 decisions, and only if they rank better on held-out decisions.

How do you avoid false alerts?

Each category-zone series has a Poisson baseline adjusted for weekday and rain. We run about 4,000 tests a day and apply one Benjamini-Hochberg correction at a 5% false-discovery rate. Today that kept 1 alert where a simple rule would have raised 15.

Can the chatbot make up numbers?

No. Counts come from read-only SQL that our compiler generates from a structured plan. The model only plans and writes words. Every number in the text is checked against the data; if one fails, it rewrites once, then uses a template. 75 out of 75 golden questions are correct with AI off.

Which AI models do you use, and why?

Amazon Bedrock Nova Lite for routing and writing (it is on the FarmwiseAI AWS account), with Gemini and Groq as backups. Local multilingual-e5 and bge models for meaning. Numbers never come from a model, which keeps it cheap, testable and safe.

What is your weakest point?

The news incident filter: F1 about 0.63, so it finds about 6 in 10 incident articles. More real labels are the fastest fix, and that is first on our roadmap. Also, linking accuracy is measured on simulated events because the department data is simulated.

What happens if a source goes down?

The Data Steward marks the feed as stale or degraded, the chip in the top bar turns orange and says which feed is behind, and the rest of the dashboard keeps working on the last good data. Endpoints closed by their owner (captcha, login) are logged, never bypassed.

Why GitHub Actions and not AWS for scheduling?

Our AWS role does not allow EventBridge scheduling, so cron-job.org triggers the GitHub job every hour. The job runs the pipeline and uploads to S3. CPCB and the Flood Monitor block cloud machines, so the team PC collects those two and sends them to the job.

Why is the website's database SQLite in memory?

We were asked to use only AWS services, with no SQL server. Each hourly build sits in S3; the site loads it into an in-memory SQLite database, which is fast and needs no server. Durable data (accounts, complaints, decisions) is written to DynamoDB through API Gateway and a Lambda.

How is personal data protected?

Mobile numbers are hashed in the pipeline, ID numbers are encrypted, passwords are bcrypt-hashed, roles are enforced on the server, real complaints are excluded from the public build files, and every Collector decision is audited.

How does adding a new source by link work?

Paste a link in Data Studio. The agent checks the host is public and robots.txt allows it, detects RSS, API, page or login, maps the fields with AI, and shows a preview. Nothing is saved until the Collector approves. Scheduled runs then use the saved mapping without AI.

How does the officer–Collector loop work?

Severe incident → the Collector approves an AI-drafted instruction to the department officer → the officer does the work and sends a completion report with photos → it appears in the Collector's My Tasks → the Collector verifies or sends it back. Every step is logged.

How do you place a news article on the map?

A Tamil-aware gazetteer of Chennai places finds mentions. A place-role step picks where the event happened, not, for example, the hospital a victim was taken to. Then point-in-polygon finds the ward. Articles that only say "Chennai" are not pinned.

How did you test the system?

About 570 automated web tests, about 100 pipeline tests and 88 news tests, a golden set of 75 chatbot questions checked by SQL, and an evaluation report each build with the accuracy of every AI step.

How would this scale to other districts?

Collectors and categories are configuration; ward boundaries, taluks and departments are reference tables. A new district needs its boundaries, its gazetteer and its department list. The pipeline and the website stay the same.

What would you do with more time?

Connect one real department system, label more real news to raise the filter's recall, collect 30 Collector decisions so priority learns, and move the PC collectors to an approved server.